IPV6

- ✓The working standard for the IPv6 protocol was published by the Internet Engineering Task Force (IETF) in 1998.
- ✓In 1997, IBM became the first commercial vendor to support IPv6 through its AIX 4.3 OS.
- ✓In 2004, Japan and Korea were acknowledged as having the first public deployments of IPv6.
- ✓The explosive growth in mobile devices including mobile phones, notebook computers, and wireless handheld devices has created a need for additional blocks of IP addresses.
- ✓IPv4 currently supports a maximum of approximately 4.3 billion unique IP addresses. IPv6 supports 340,282,366,920,938,463,463,374,607,431,768,211,456 .

  340-Undecillion 282-decillion 366 nonillion 920-octillion 938-septillion 463-sextillion 463-quintillion 374-quadrillion 607-trillion 431 billion 768-million 211-thousand 456

  3.6 million IPs for 1 square feet
- ✓Recent advancements in network technology including Network Address Translation (NAT) have temporarily lessened the urgency for new IP addresses.

IPV6 Advantages

- ✓ Large address space
- ✓ Global reachable and flexibility - Access computer from anywhere
- ✓ Aggregation - summarization
- ✓ Multi-homing - multiple internet connection to same PC
- ✓ Stateless Auto configuration - DHCP
- ✓ Plug and Play
- ✓ IPV5 (Internet stream protocol)
- ✓ Renumbering - assigning IP manually
- ✓ No need of NAT & PAT
- ✓ Header is much simpler than IPV4
- ✓ No broadcast traffic, they doesn't use any layer 3 broadcasts. No ARP, they use NDP
- ✓ Unicast ,Multicast and Anycast
- ✓ Mobile IP – e.g. Roaming facility in sim (same ip can be used in different places)
- ✓ IPSec native (default, Mandatory)
- ✓ Migration tools: IPv4 and IPv6 are not compatible with each other so we need tools (Dual stack , Tunneling & NAT-PT)

IPV6 ADDRESS

2031:0000:0000:013F:0000:0000:0000:0001

2031::13F:0:0:0:1

Use : : only once

2031:0:0:13F::1

1080:0000:0000:0000:0000:0034:0000:417A

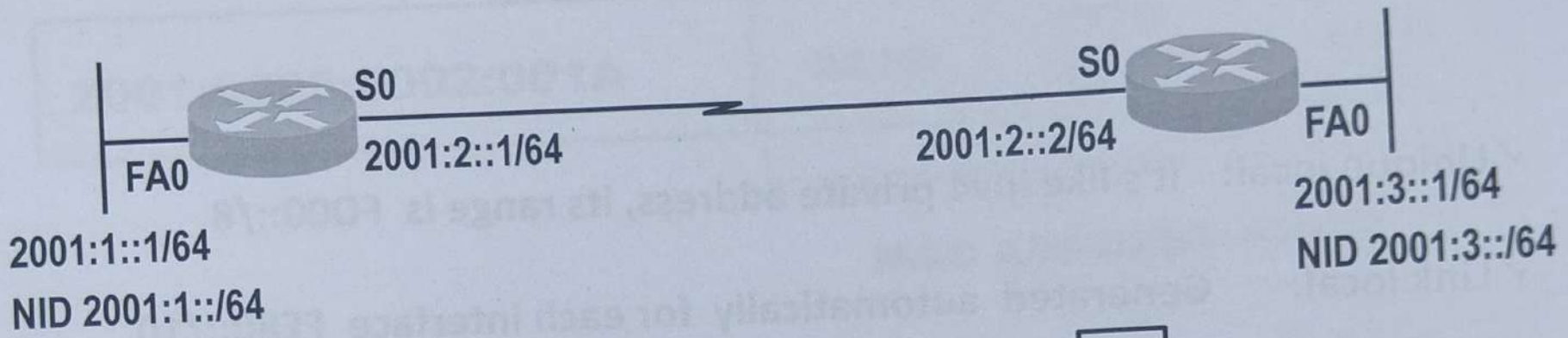
1080:0:0:0:0:34:0:417A

1080::34:0:417A

IPV6 ADDRESS TYPES

- ✓Unique local: It's like ipv4 private address, its range is FD00::/8
- ✓Link local: Generated automatically for each interface FE80::/10
- ✓Global unicast: It's like public address, its address space is 2000::/3
- ✓Unspecified: Shows as ::/128 when host has no usable IPV6 address
- ✓Loopback: Same as 127.0.0.1 in IPV4, in IPV6 ::1/128
- ✓Multicast: Reserved space for this is FF::/8
- ✓Tunneling: 2002::/16 range reserved for tunneling

IPV6 Static routing



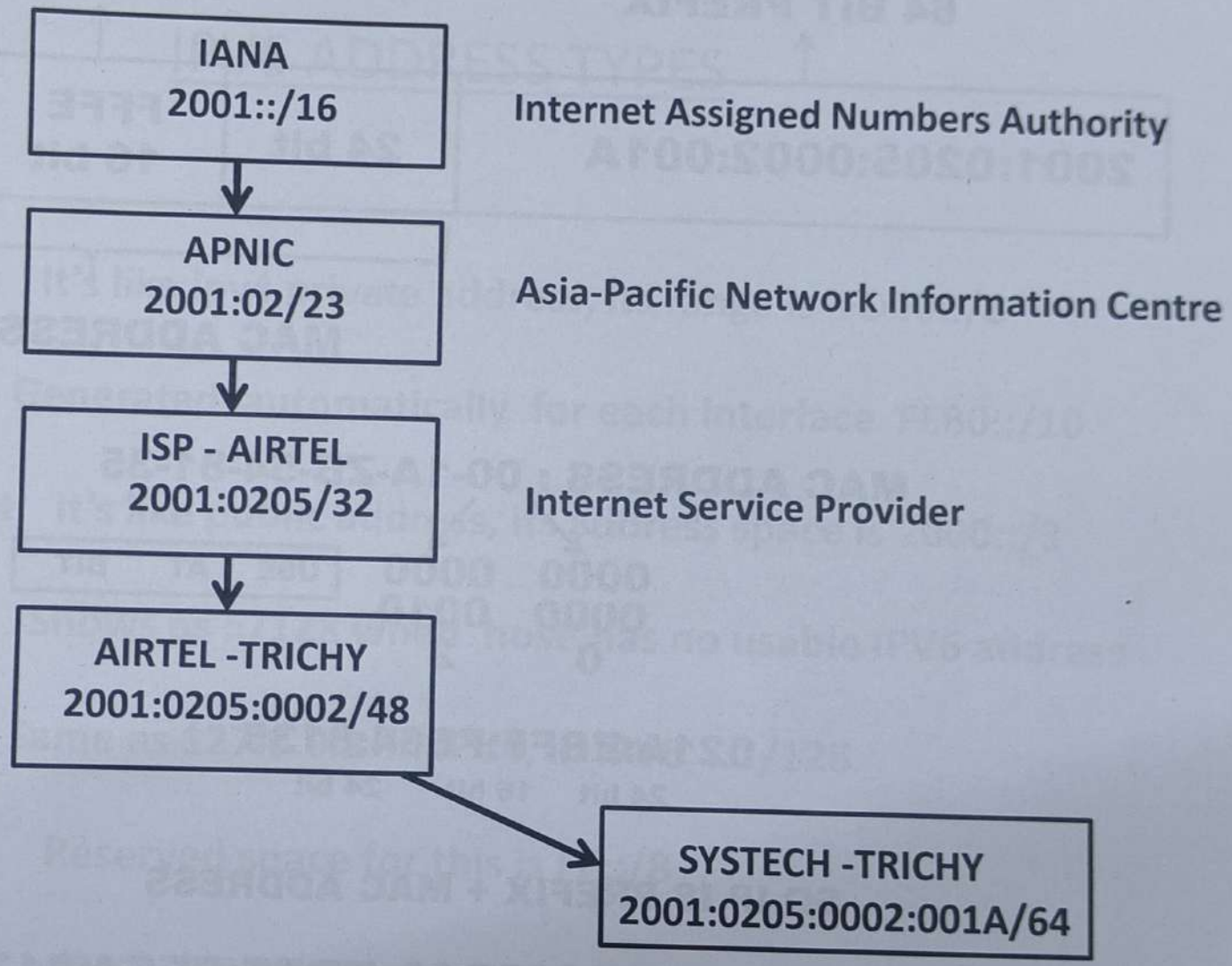
RA

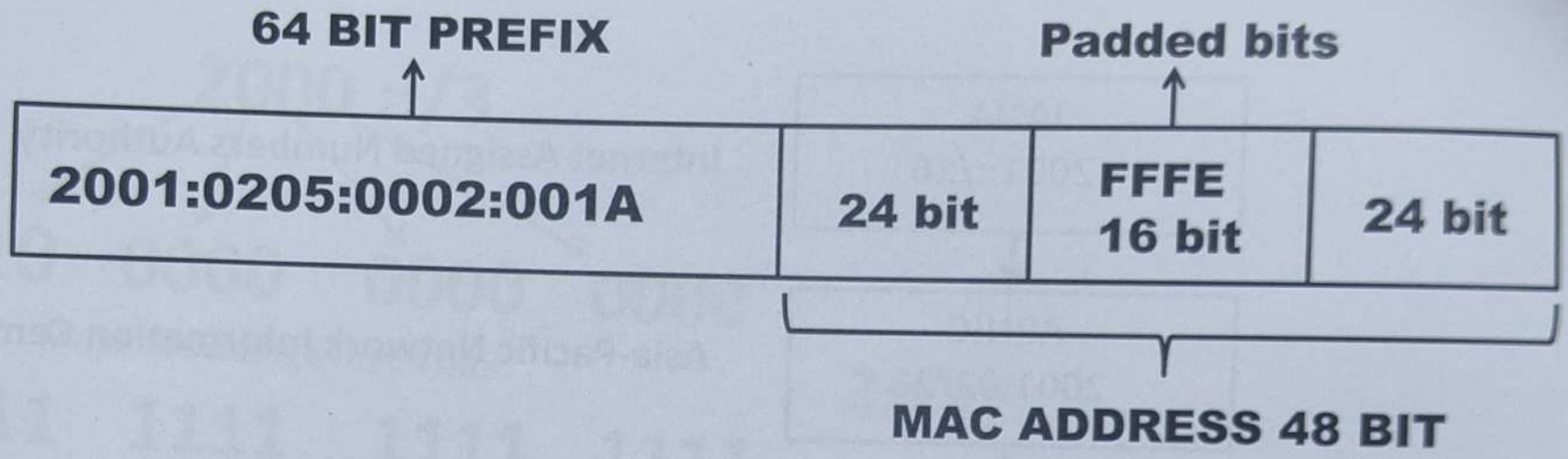
```
#int fa0
#ipv6 add 2001:1::1/64
#no sh
#int s0
#ipv6 add 2001:2::1/64
#no sh
#exit
#ipv6 unicast-routing
#ipv6 route 2001:3::/64  2001:2::2
```

RB

```
#int fa0
#ipv6 add 2001:3::1/64
#no sh
#int s0
#ipv6 add 2001:2::2/64
#no sh
#exit
#ipv6 unicast-routing
#ipv6 route 2001:1::/64  2001:2::1
```

```
#sh ipv6  interface brief
#sh ipv6  route
```


IANA
2001::/16
Internet Assigned Numbers Authority
APNIC
2001:02/23
Asia-Pacific Network Information Centre
ISP - AIRTEL
2001:0205/32
Internet Service Provider
AIRTEL -TRICHY
2001:0205:0002/48
SYSTECH -TRICHY
2001:0205:0002:001A/64



MAC ADDRESS : 00-1A-ZB-54-81-35

0000 0000 0	0000 0010 2	USE 1 AT BIT

021A:ZBFF:FE54:8135

24 bit 16 bit 24 bit

SO IP IS PREFIX + MAC ADDRESS

2001:0205:0002:001A:021A:ZBFF:FE54:8135/128BIT

#ipv6 address 2001:205:2:1A/64 eui-64

IPV6 RIPNG (Next Generation)

RIP V2 (IPV4)		RIPNG (IPV6)
UDP port	520	521
Auto-Summary	Possible	Unavailable
Multicast address	224.0.0.9	FF02::9

RA & RB

```
#ipv6 unicast-routing
#ipv6 router  rip systech
#exit
#int fa0
#ipv6 rip systech enable
#exit
#int s0
#ipv6 rip systech enable
#exit
#sh ipv6  route
```

2000 ::/3

0010 0000 0000 0000

Must not change first 3 bits

0011 1111 1111 1111

3 F F F

2000:: 3FFF::

FE80 ::/10

1111 1110 1000 0000

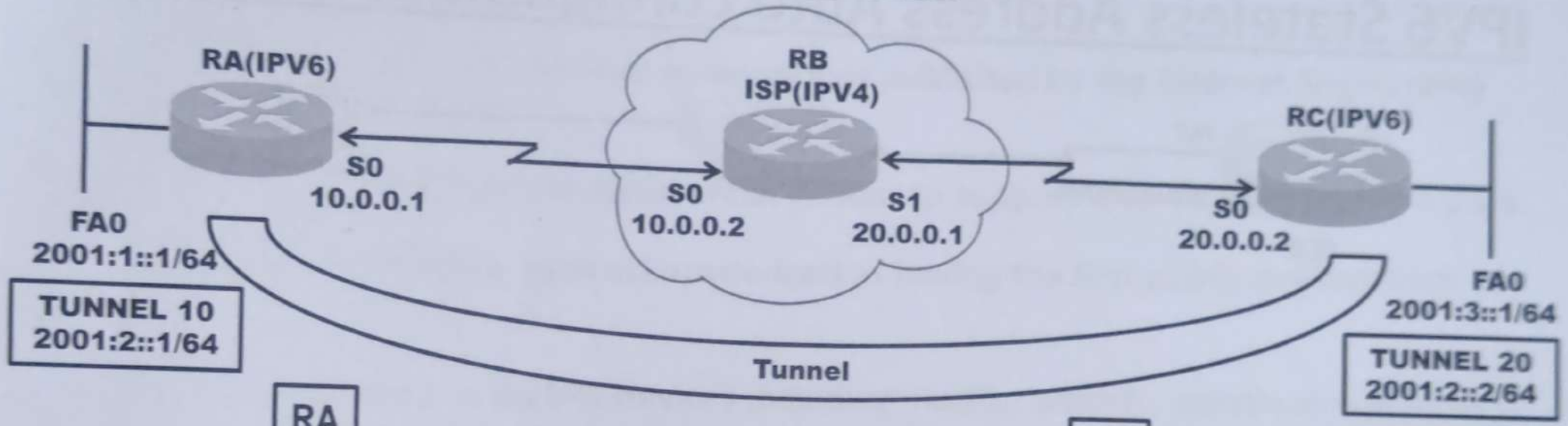
Must not change first 10 bits

1111 1110 1011 1111

F E B F

FE80:: ..... FEBF::

IPV6 Tunnel



RA

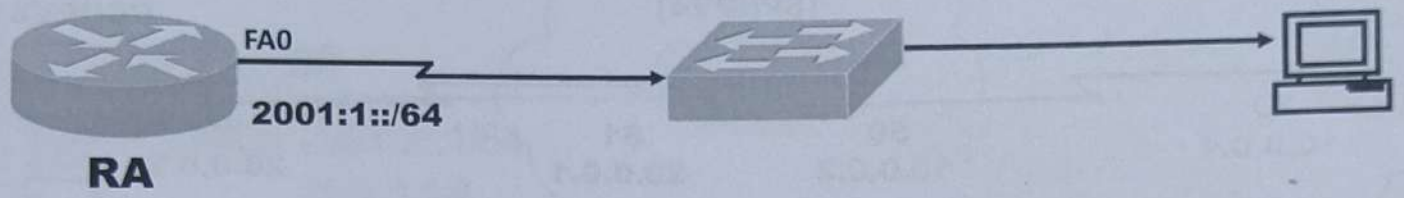
```
# interface tunnel 10
# ipv6 add 2001:2::1/64
# tunnel source s0
# tunnel  destination 20.0.0.2
# tunnel mode ipv6ip
# exit
# ipv6 unicast-routing
# ipv6 route 2001:3::/64 2001:2::2
# ip route 20.0.0.0 255.0.0.0  10.0.0.2
```

RC

```
# interface tunnel 20
# ipv6 add 2001:2::2/64
# tunnel source s0
# tunnel  destination 10.0.0.1
# tunnel mode ipv6ip
# exit
# ipv6 unicast-routing
# ipv6 route 2001:1::/64 2001:2::1
# ip route 10.0.0.0 255.0.0.0  20.0.0.1
```

```
#sh ipv6 route
```

IPV6 Stateless Address Auto configuration (SLAAC)



RA # ipv6 unicast-routing

RA # interface fa0

RA # ipv6 address 2001:1::/64 eui-64

RA # debug ipv6 nd

IPV6 OSPF

OSPF (IPV4)		OSPF (IPV6)
Multicast all OSPF	224.0.0.5	FF02::5
Multicast DR/BDR	224.0.0.6	FF02::6

RA&RB

```
#ipv6 unicast-routing
#ipv6 router  ospf 10
#router –id 10.10.10.10
#exit
#int fa0
#ipv6 ospf 10 area 0
#exit
#int s0
#ipv6 ospf 10 area 0
#exit
```

```
#sh ipv6  route
#sh ipv6 ospf neighbors
#sh ipv6 ospf database
```

IPV6 EIGRP

EIGRP (IPV4)		EIGRP (IPV6)
Protocol header	88	88
Auto-Summary	Possible	Unavailable
Multicast address	224.0.0.10	FF02::10

RA&RB

```
#ipv6 unicast-routing
#ipv6 router  eigrp 10
#no sh
#router –id 10.10.10.10
#exit
#int fa0
#ipv6 eigrp 10
#exit
#int s0
#ipv6 eigrp 10
#exit
```

```
#sh ipv6 route
#sh ipv6 eigrp neighbors
#sh ipv6 eigrp topology
```

IP Dual Stack

```
#int fa0
#ipv6 add 2001:1::1/64
#ip add 10.0.0.1 255.0.0.0
#no sh
#int s0
#ipv6 add 2001:2::1/64
#ip add 20.0.0.1 255.0.0.0
#no sh
#exit
#ip route 30.0.0.0 255.0.0.0 20.0.0.2
#ipv6 unicast-routing
#ipv6 route 2001:3::/64  2001:2::2
```

```
#int fa0
#ipv6 add 2001:3::1/64
#ip add 30.0.0.1 255.0.0.0
#no sh
#int s0
#ipv6 add 2001:2::2/64
#ip add 20.0.0.2 255.0.0.0
#no sh
#exit
#ip route 10.0.0.0 255.0.0.0  20.0.0.1
#ipv6 unicast-routing
#ipv6 route 2001:1::/64  2001:2::1
```

```
#sh ipv6  route
```